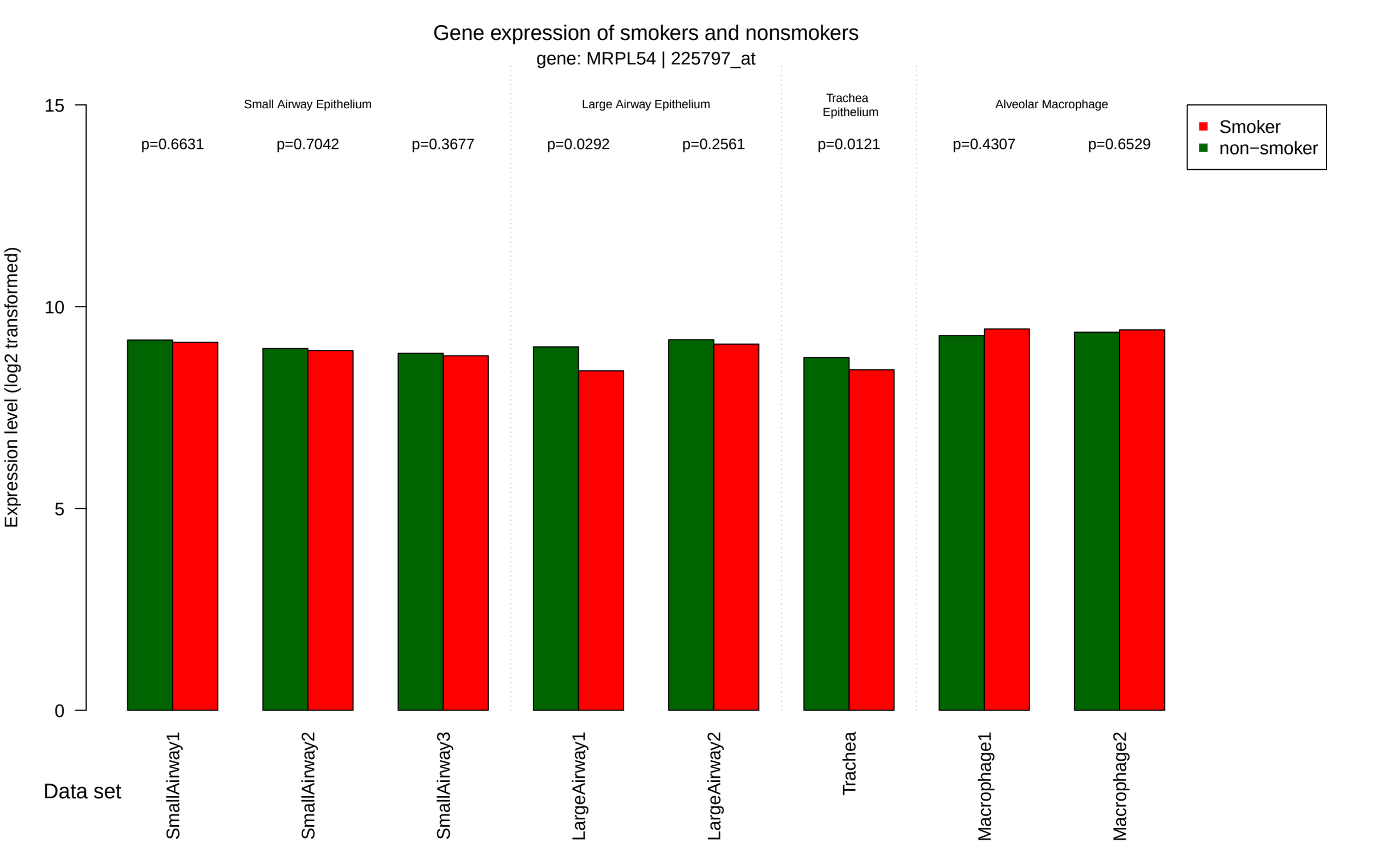
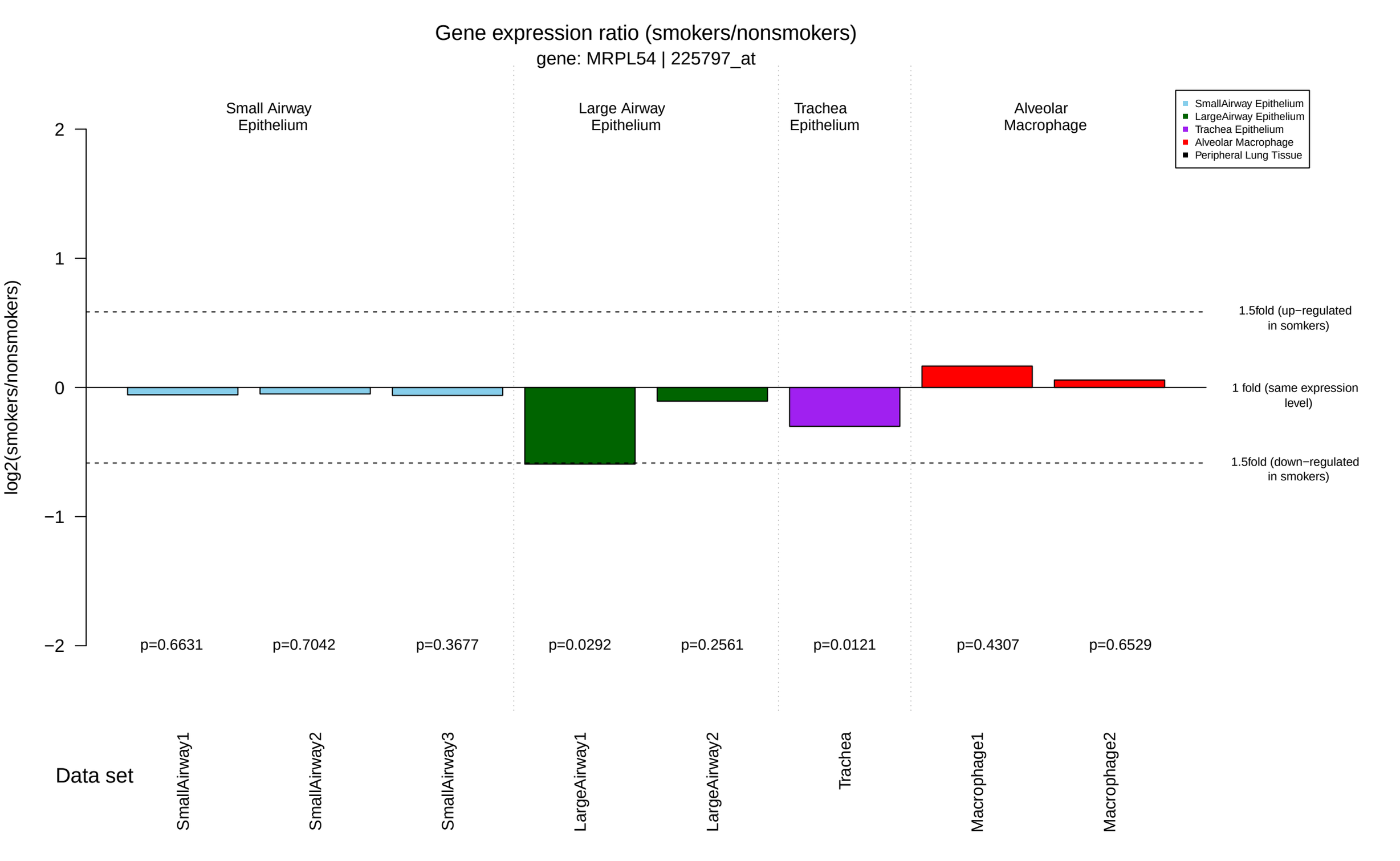
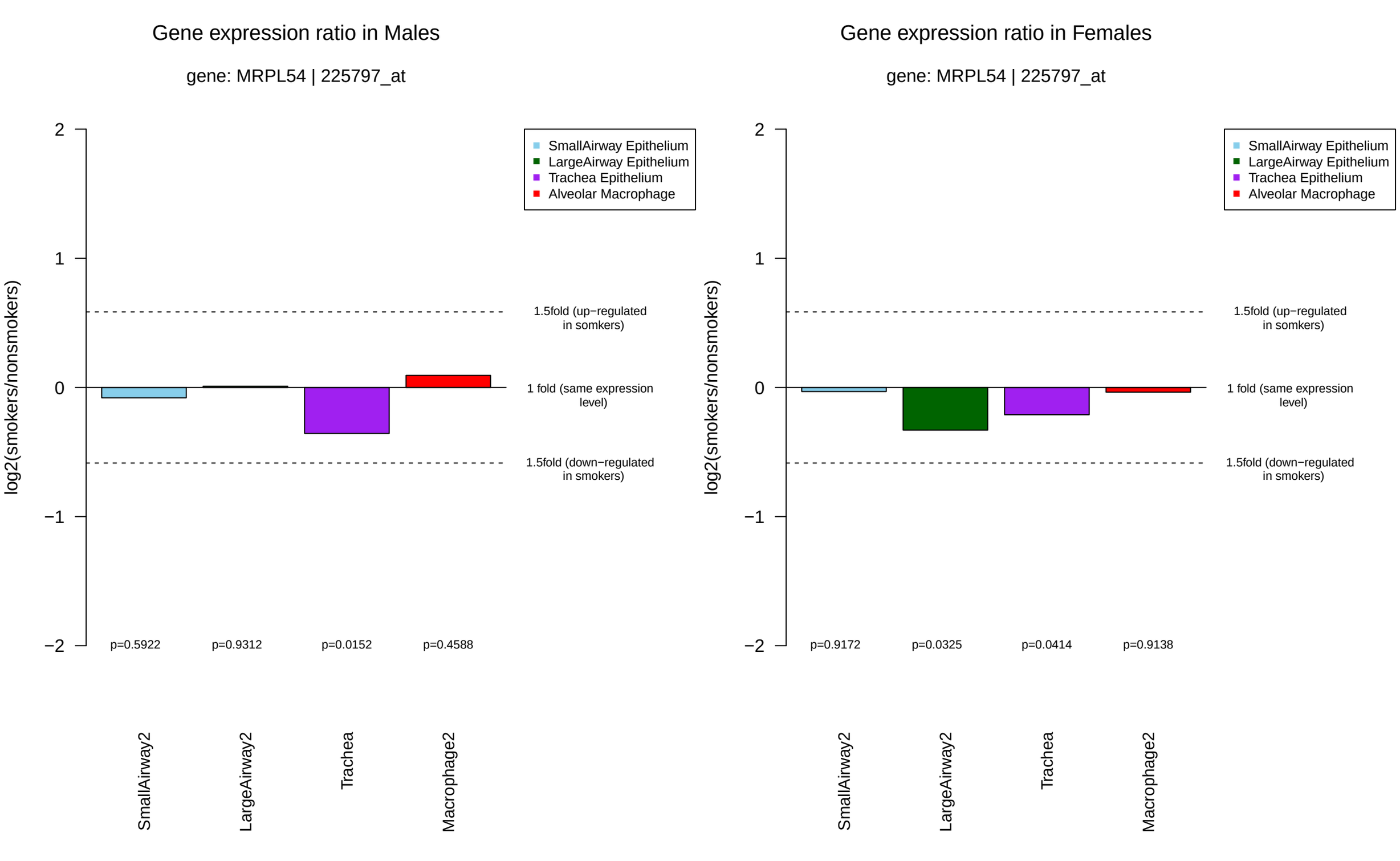
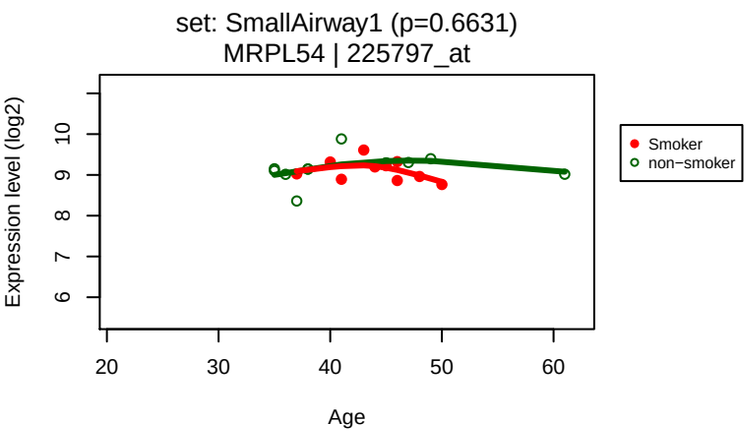
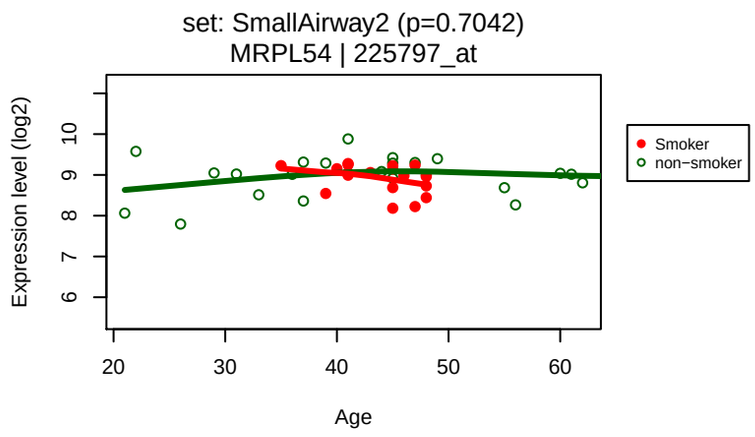
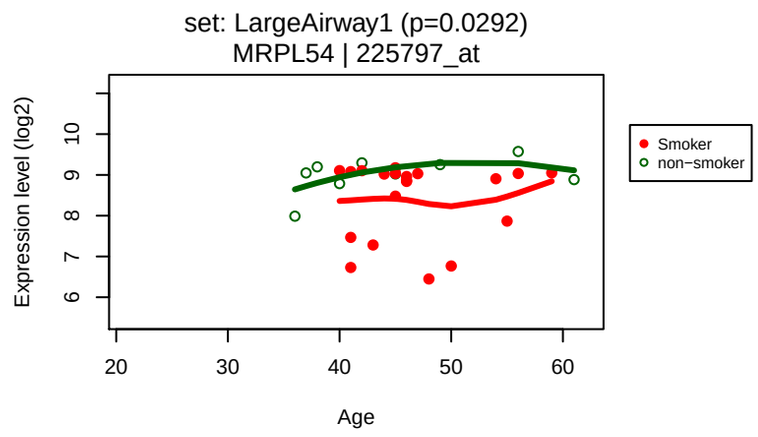

Gene expression of smokers and nonsmokers
gene: MRPL54 | 225797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6631
p=0.7042
p=0.3677
p=0.0292
p=0.2561
p=0.0121
p=0.4307
p=0.6529
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

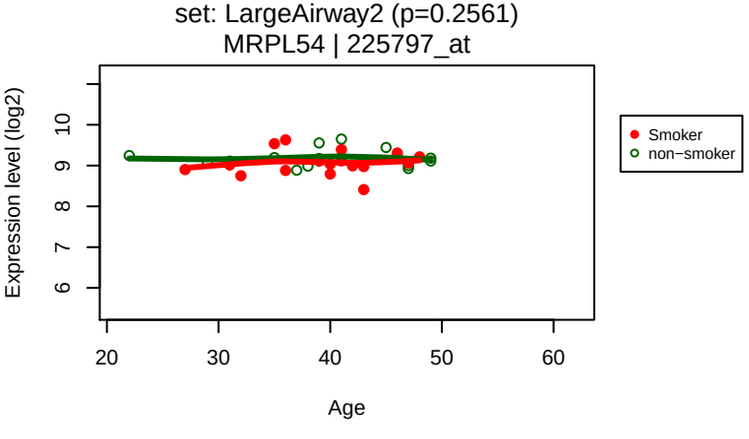

Gene expression ratio (smokers/nonsmokers)
gene: MRPL54 | 225797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6631
p=0.7042
p=0.3677
p=0.0292
p=0.2561
p=0.0121
p=0.4307
p=0.6529
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

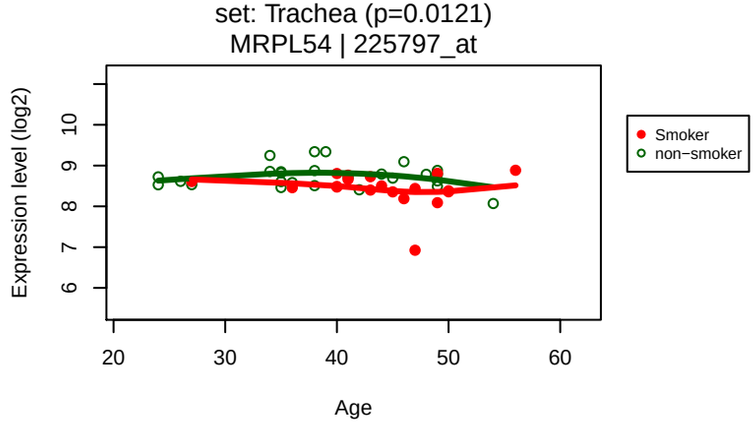

Gene expression ratio in Males
gene: MRPL54 | 225797_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5922
p=0.9312
p=0.0152
p=0.4588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL54 | 225797_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9172
p=0.0325
p=0.0414
p=0.9138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

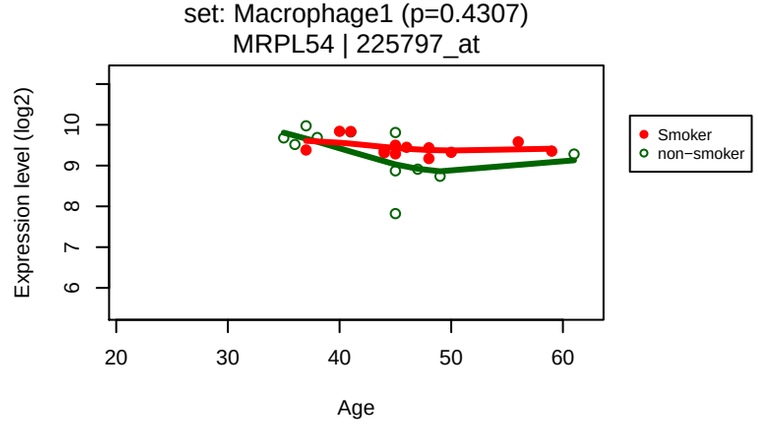

set: SmallAirway1 (p=0.6631)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker

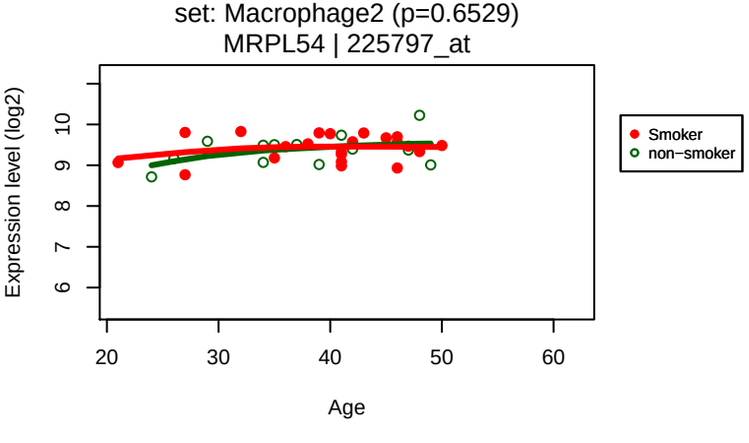

set: SmallAirway2 (p=0.7042)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker

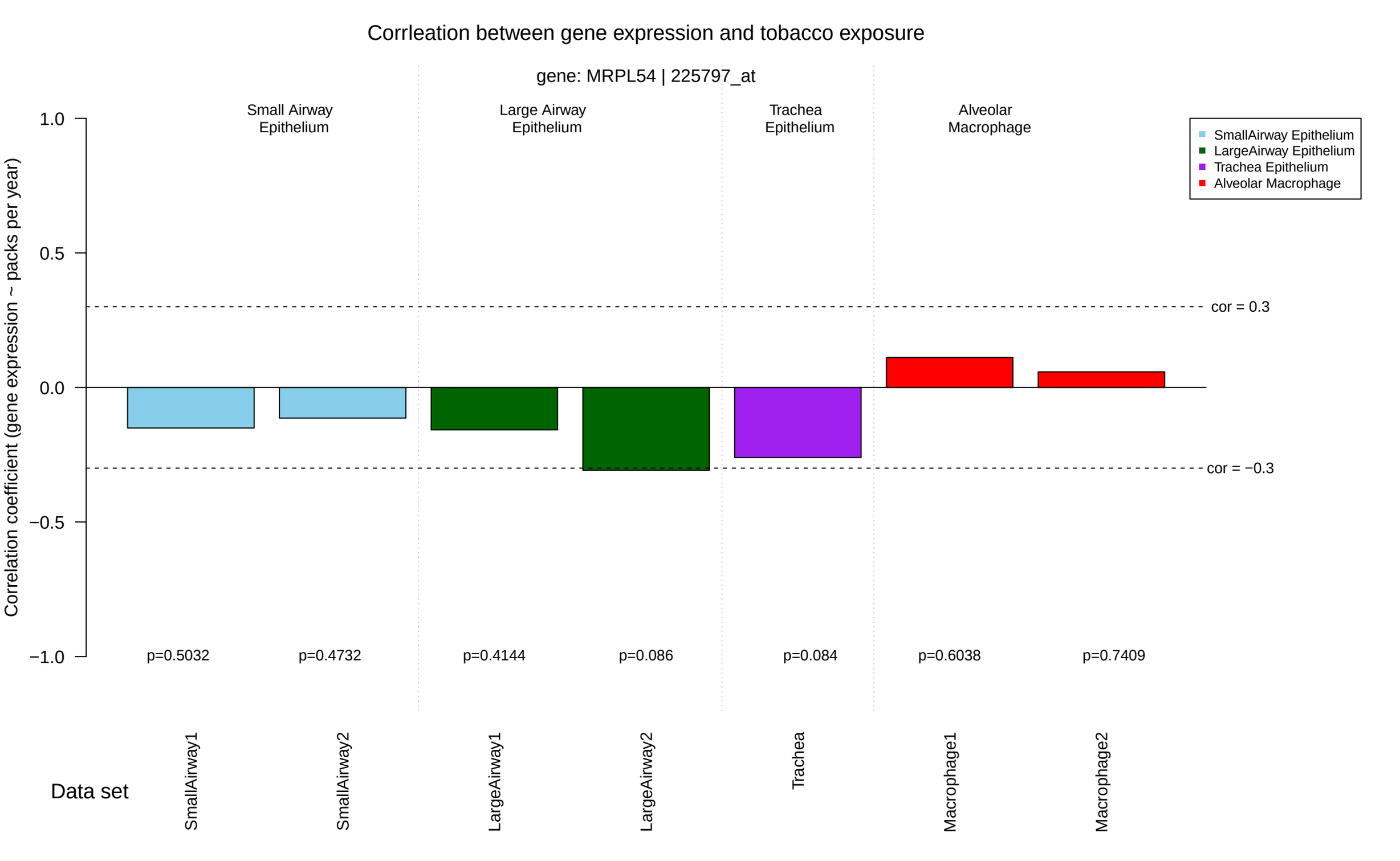

set: LargeAirway1 (p=0.0292)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2561)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0121)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4307)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6529)
MRPL54 | 225797_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL54 | 225797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5032
p=0.4732
p=0.4144
p=0.086
p=0.084
p=0.6038
p=0.7409
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2